Crystal Field theory to explain observed properties of complexes: Variation of some physical properties across a period:

1. Lattice energy of transition metal ions in a complex
2. Ionic radii of transition metal ions in a complex
3. Enthalpy of hydration of transition metal ions
4. Site preference of Spinels and Inverse spinels

Lattice Energy: ***Energy released when one mole of an ionic solid is formed from isolated gaseous ions***

Calculated theoretically using the Born- Lande Equation

$$U_o = \frac{A N Z^+ Z^- e^2}{4\pi\varepsilon_o r_o}\left(1 - \frac{1}{n}\right)$$

Where

A = Madelung constant (related to the geometry of the crystal)

N = Avogadro's number

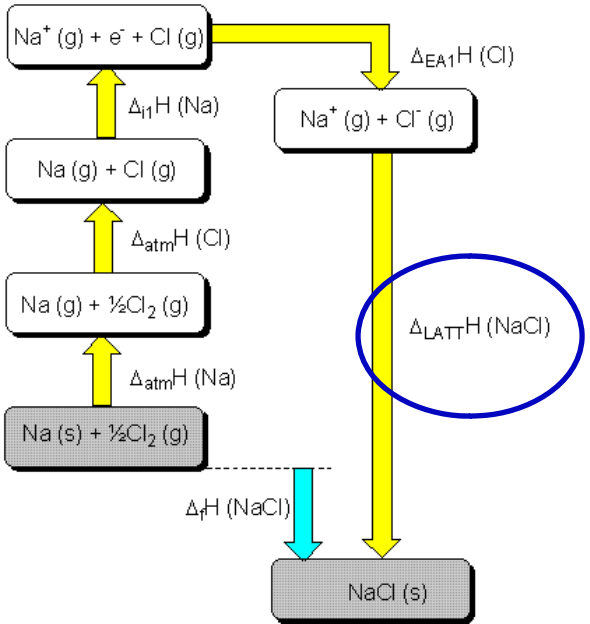
Z= Charge on the M+ and M- ions

ε_0= permittivity of free space

r_0 = distance to the closest ion

n= Born exponent (a number between 5 and 12)

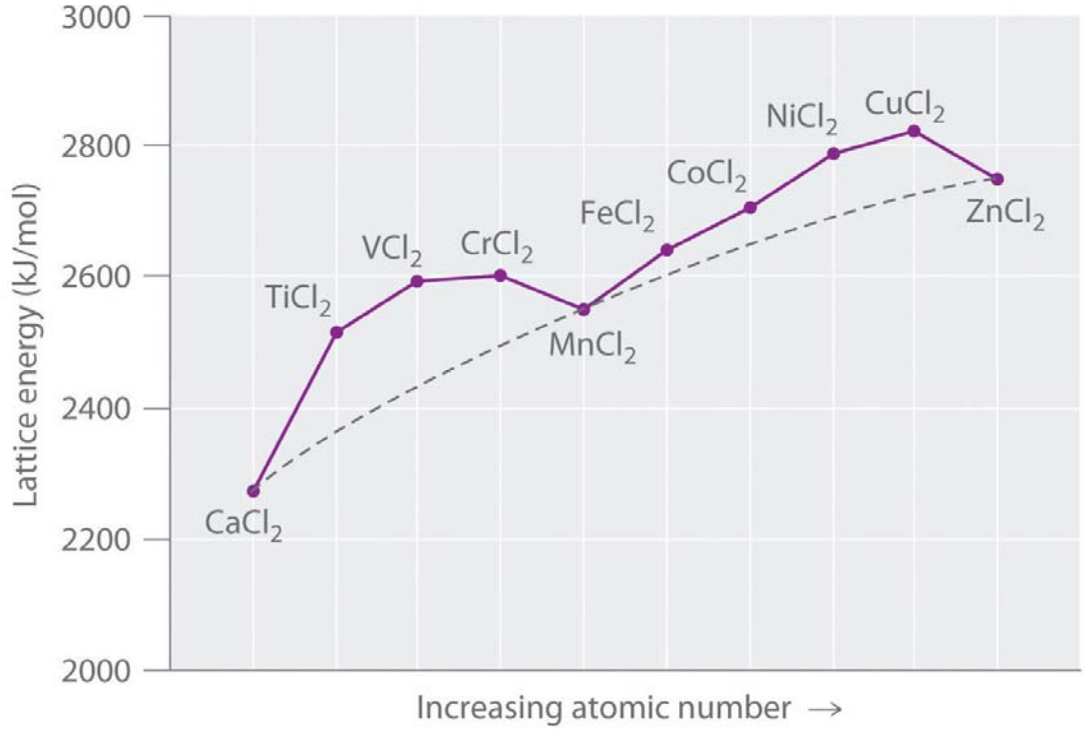
Experimentally determined using the Born- Haber cycle

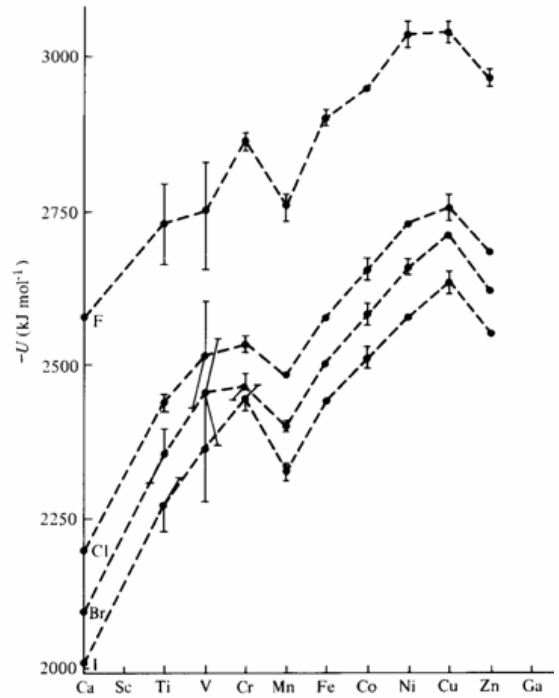


Lattice Energies (1st Row Divalent Transition Metal Halides and Main group halides)

Compound	Lattice energy (KJ/mol)		Differences
	Measured	Calculated	(KJ/mol)
NaCl	-764	-764	0
MgF_2	-2908	-2915	+7
MnF_2	-2770	-2746	+24
FeF_2	-2912	-2752	-160
CuF_2	-3042	-2885	-157
NiF_2	-3046	-2917	-129

The discrepancy for a six- coordinate transition metal ion can be explained by CFSE





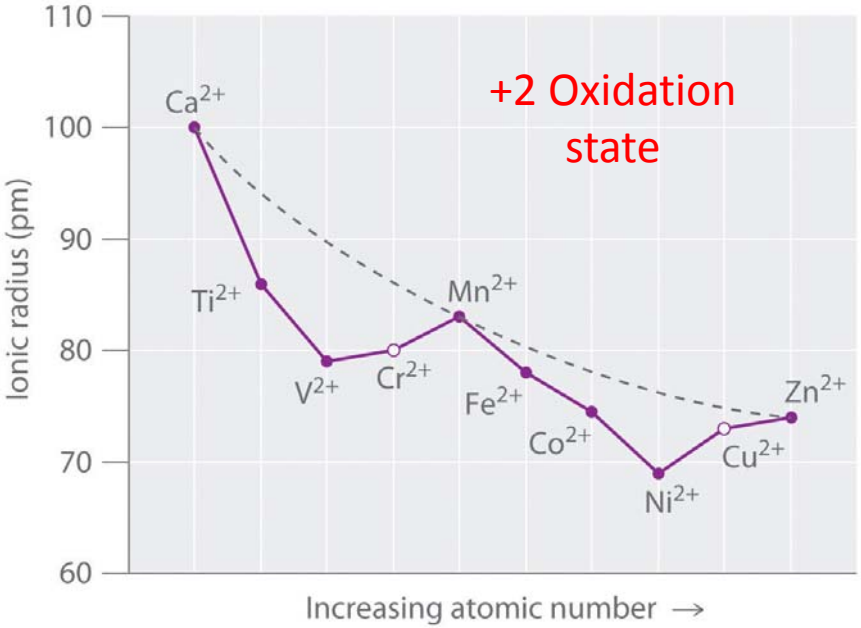
- According to the Born –Lande Equation one can expect a smooth increase in lattice energies as we go from left to right due to decrease in ionic radius of the metal ions
- As anticipated a smooth curve is not seen: instead a double hump shaped curve is obtained
- Ca^{2+} (d^0), Mn^{2+}(d^5 HS) and Zn^{2+} (d^{10}) which in common have CFSE =0 lie almost on the expected line.
- Ions such as V^{2+} which show high CFSE in a weak field situation with high lattice energy values show significant deviation from the calculated lattice energies

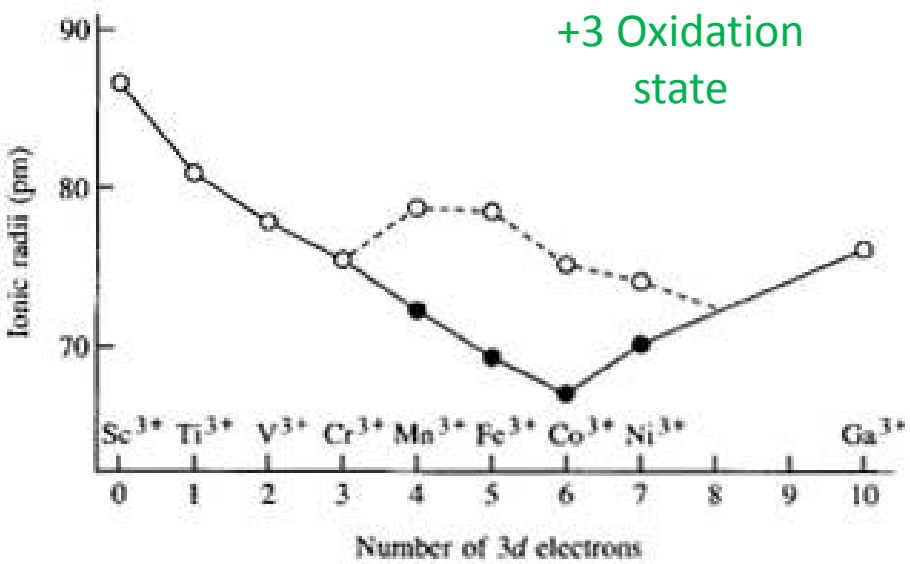
Ionic radii of transition metal ions in the presence of weak field and strong field ligands

One can expect decrease the ionic radii of the M^{2+} ions smoothly from Ca^{2+} to Zn^{2+} due to the increase in nuclear charge

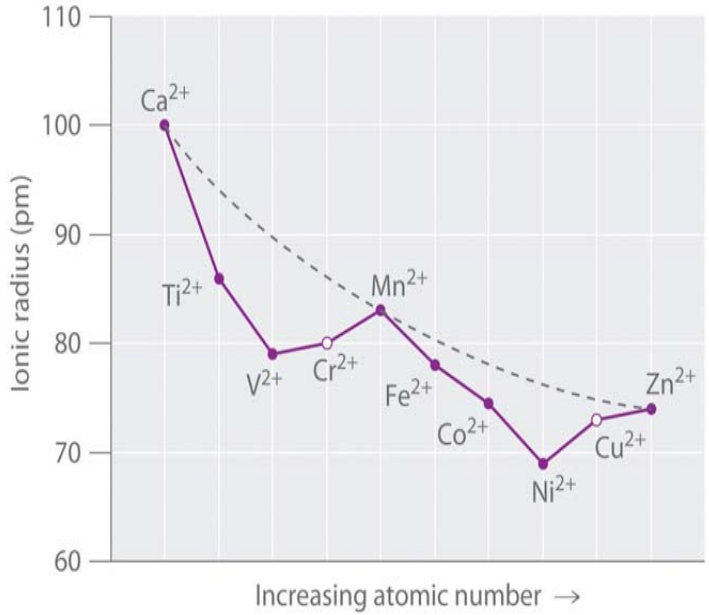
But the plot shown below (left) for weak field ligands indicate that the expected regular decrease is absent expect for Ca^{2+}, Mn^{2+} and Zn^{2+}

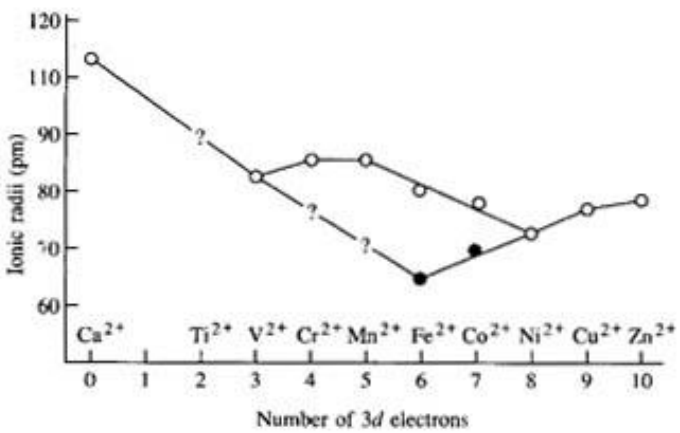
For strong field ligands like CN- a different trend in variation is observed with a steady decrease till d^6 (t_{2g}^6)





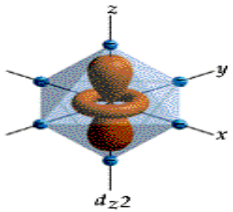
Ionic radii of transition metal ions in the presence of weak field and strong field ligands



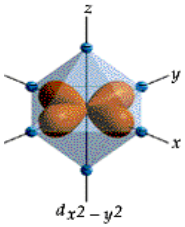


Why does the ionic radii decreases and then increases??

Ti^{2+} (d^2) electron occupy only t_{2g}
V^{2+} (d^3) electrons occupy only t_{2g}
Cr^{2+} (d^4HS) electrons start occupying the e_g orbitals



As the e_g orbitals point directly towards the ligands, the repulsion between the metal electrons and ligand electrons will be higher than normal leading to the eventual increase in the ionic radius.



In the case of strong field ligand such as cyanide there will be a steady decrease in ionic radii till t_{2g}^6 is reached.

The same tend is observed also for M^{3+} transition metal complexes

Variation of some physical properties across a period

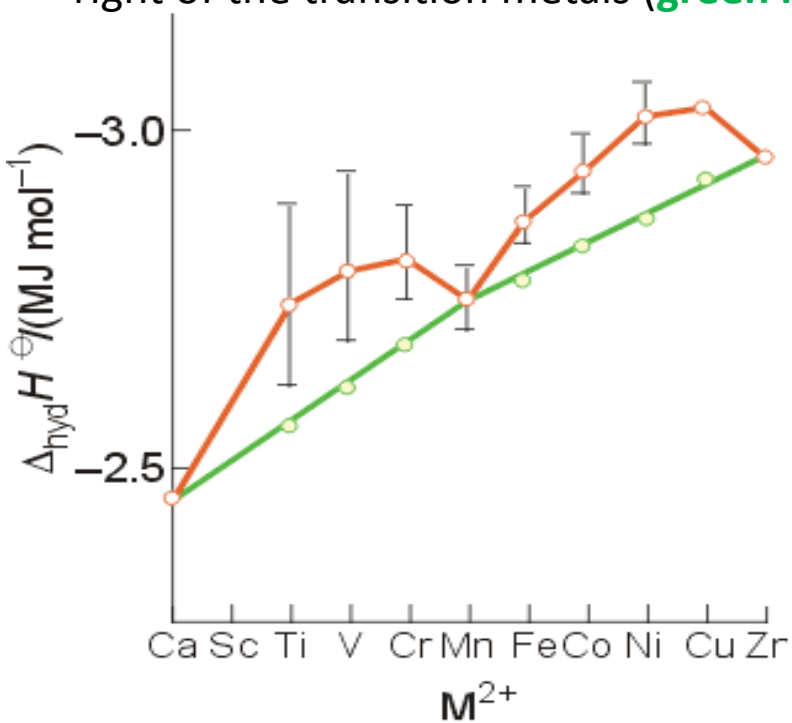
Enthalpy of hydration of transition metal ions

It is the heat exchange involved when 1 mole of gaseous ions become hydrated

$M_{(g)}^{2+}$ + excess H_2O ⟶ $[M(H_2O)_6]^{2+}$

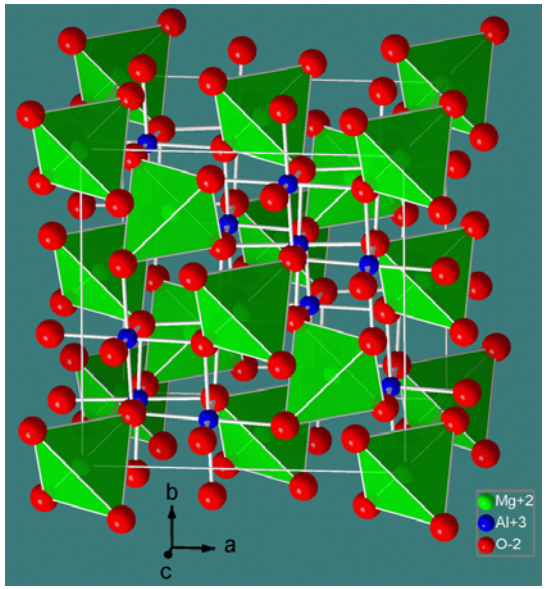
The amount of energy released when a mole of the ion dissolves in a large amount of water forming an infinite dilute solution in the process.

Higher the charge on the ions and smaller the size, more exothermic will be the hydration energy. So it is expected to increase smoothly on going from left to right of the transition metals (**green line in the graph**)



The heats of hydration show two "humps" consistent with the expected CFSE for the metal ions. The values for d^5 and d^{10} are the same as expected with a CFSE equal to 0.

Site preference of Spinels and Inverse Spinels

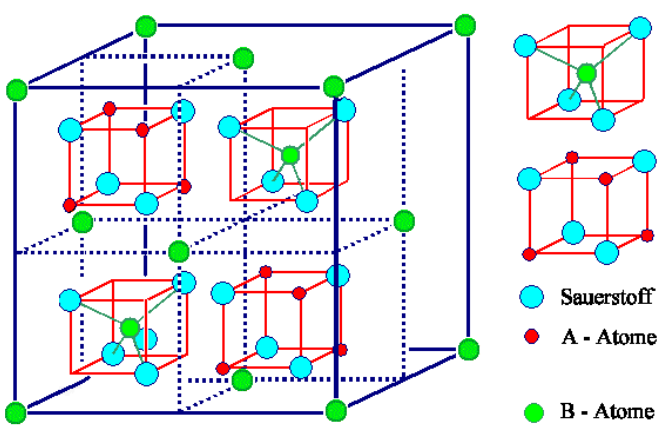


Spinels are a class of crystalline solids of the general formula AB_2O_4 ($A^{II}B^{III}{}_2O_4$) where

A = main group (Group IIA) or transition metal ion in the +2 oxidation state

B= main group (Group IIIA) or transition metal ion in the +3 oxidation state

The weak field oxide ions provide a cubic close-packed lattice. In one unit cell of AB_2O_4 there are 8 tetrahedral and 4 octahedral holes



AB_2O_4 Spinell Die roten "Würfel" sind auch im hinterenTeil des Kristalls

Normal Spinel: A^{2+} ions occupy 1/8th of the Td holes
B^{3+} ions occupy ½ of the octahedral holes

Inverse Spinel: The +2 ion and one of the +3 ions exchange positions
i.e. Td holes: +3 ,
Oh holes: both +2 and +3 ions

Eg. $MgAl_2O_4$

Mg2+ Tetrahedral
Al3+ octahedral

Eg. Fe_3O_4 (Magnetite)

Fe3+ Tetrahedral
Fe 2+, Fe3+ octahedral

If A and B of AB_2O_4 are both s or p block elements (e.g. $MgAl_2O_4$) it always show Spinel structure

Question: **Why does some AB_2O_4 compounds having transition elements as A and /or B prefer the inverse Spinel structure and some others normal Spinel structure?**

ANS: Crystal Field Stabilization Energy in an octahedral site [Also given as Octahedral site stabilization energy (OSSE) in some books]

$Mn_3O_4 = Mn^{II}Mn^{III}{}_2O_4$

O^{2-} = a weak field ligand
$Mn^{2+} = d^5$ HS : CFSE = 0
$Mn^{3+} = d^4$ HS : CFSE = -0.6 Δ_o

Mn^{2+} by exchanging positions with Mn^{3+} in an octahedral hole is not going to gain any extra crystal field stabilization energy. While Mn^{3+} by being in the octahedral hole will have CFSE. Therefore Mn_3O_4 will be Normal Spinel

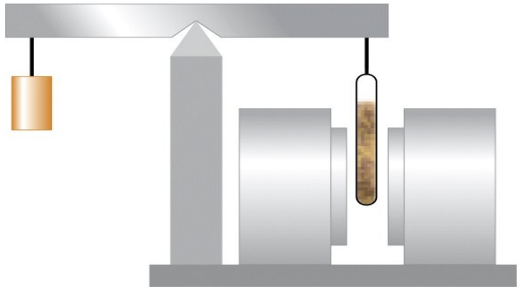
$Fe_3O_4 = Fe^{II}Fe^{III}{}_2O_4$

$Fe^{2+} = d^6$ HS : CFSE =-0.4 Δ_o
$Fe^{3+} = d^5$ HS: CFSE = 0

Fe^{2+} by exchanging positions with Fe^{3+} to an octahedral hole is going to gain extra crystal field stabilization energy. While Fe^{3+} by being in the octahedral hole will not have any CFSE. Therefore Fe_3O_4 will be Inverse Spinel

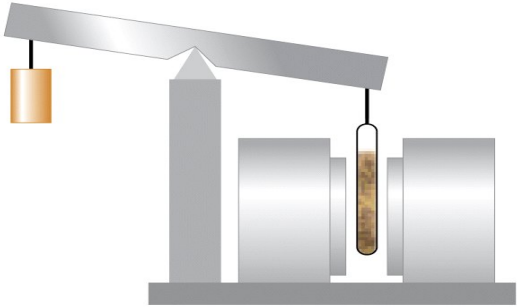
Magnetic properties of transition metal complexes

Magnet off



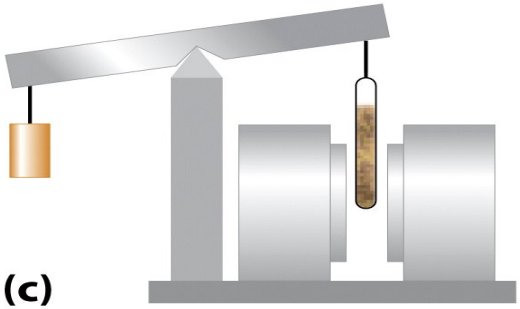
(a)

Magnet on:
Paramagnetic

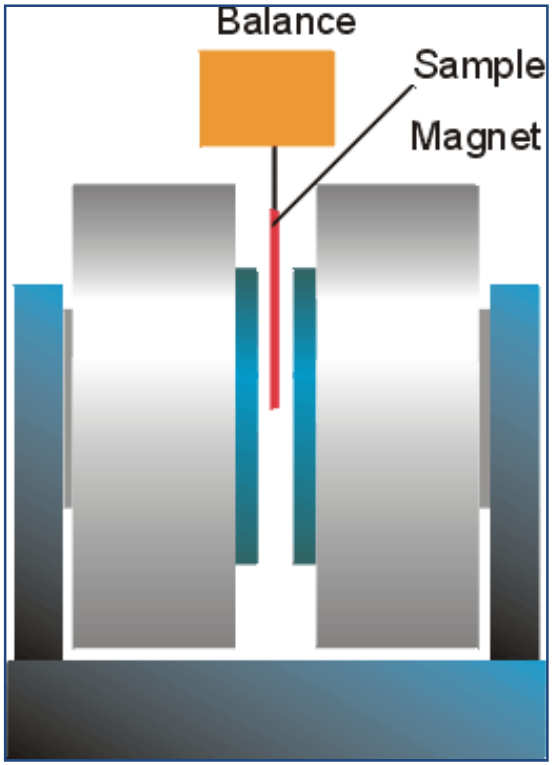


(b)

Magnet on:
diamagnetic



(c)

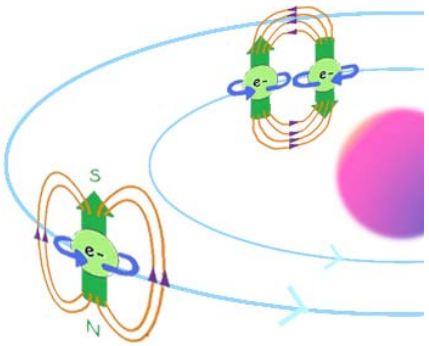


Gouy balance to measure the magnetic susceptibilities

Magnetic properties of metal complexes

Magnetism is caused by moving charged electrical particles (Faraday, 1830s). These particles can be the current of electrons through an electric wire, or the movement of charged particles (protons and electrons) within an atom. These charged particles move much like planets in a solar system:

- ***nucleus spin*** around its own axis, causing a **very weak** magnetic field.
- *electrons orbit* around the nucleus, causing a **weak** magnetic field.
- *electrons spin* around their own axis, causing a **significant** magnetic field .



Spinning electrons generate the bulk of the magnetism in an atom.

- Within each orbit, electrons with opposite spins pair together, resulting in no net magnetic field. Therefore **only unpaired electrons** lead to magnetic moment

The **spin-only** formula (μ_s)

$$\mu_S = \sqrt{n(n+2)}$$

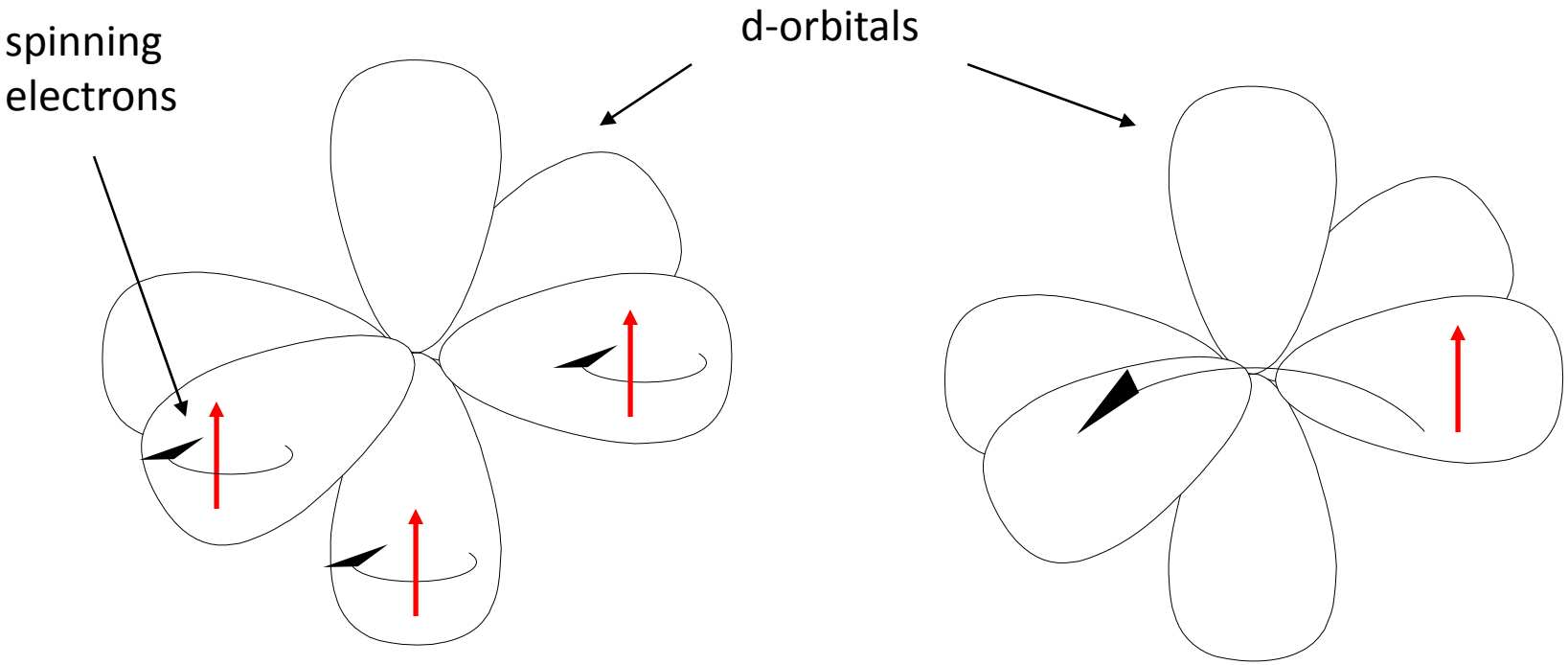
Magnetic properties: Spin only and effective

The **spin-only** formula (μ_s) applies reasonably well to metal ions from the first row of transition metals: (units = μ_B, Bohr-magnetons)

Metal ion	d^n configuration	μ_s(calculated)	μ_{eff} (observed)
Ca^{2+}, Sc^{3+}	d^0	0	0
Ti^{3+}	d^1	1.73	1.7-1.8
V^{3+}	d^2	2.83	2.8-3.1
V^{2+}, Cr^{3+}	d^3	3.87	3.7-3.9
Cr^{2+}, Mn^{3+}	d^4	4.90	4.8-4.9
Mn^{2+}, Fe^{3+}	d^5	5.92	5.7-6.0
Fe^{2+}, Co^{3+}	d^6	4.90	5.0-5.6
Co^{2+}	d^7	3.87	4.3-5.2
Ni^{2+}	d^8	2.83	2.9-3.9
Cu^{2+}	d^9	1.73	1.9-2.1
Zn^{2+}, Ga^{3+}	d^{10}	0	0

Spin and orbital contributions to μ_{eff}

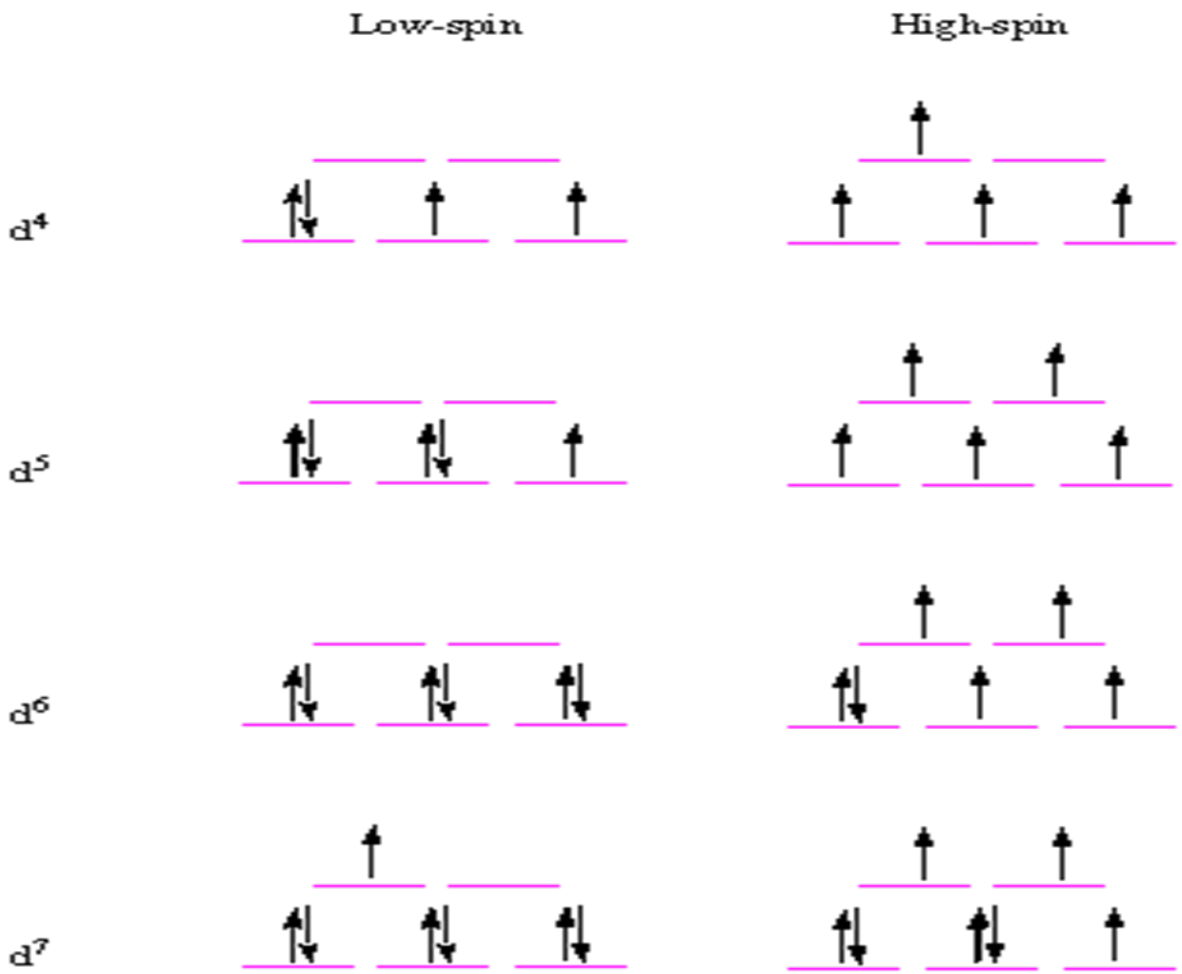
For the first-row d-block metal ions the main contribution to magnetic susceptibility is from electron spin. However, there is also an <u>orbital contribution</u> (especially for the second and third row TM) from the motion of unpaired electrons from one d-orbital to another. This motion constitutes an electric current, and so creates a magnetic field.



spin contribution – electrons are spinning creating an electric current and hence a magnetic field

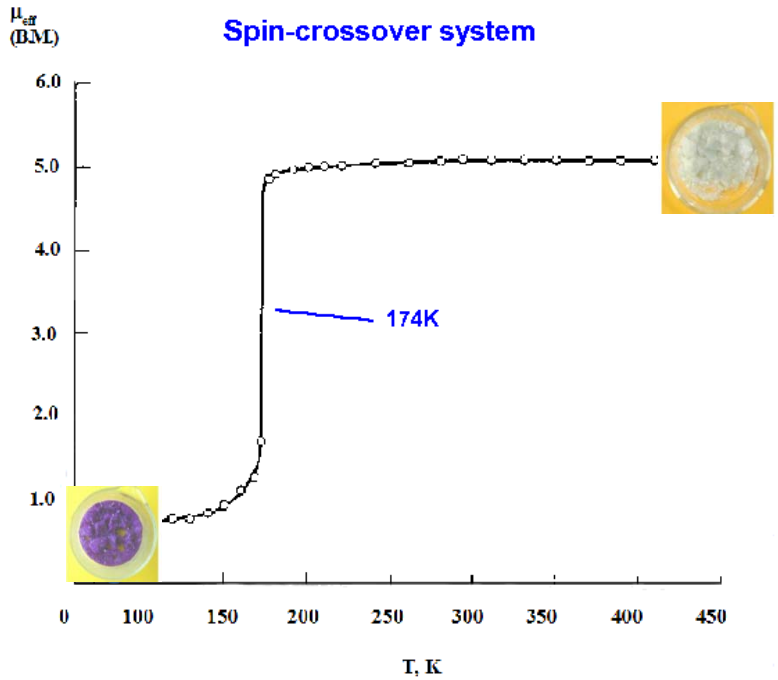
orbital contribution - electrons move from one orbital to another creating a current and hence a magnetic field

Crystal Field theory explains magnetic moments of TM complexes

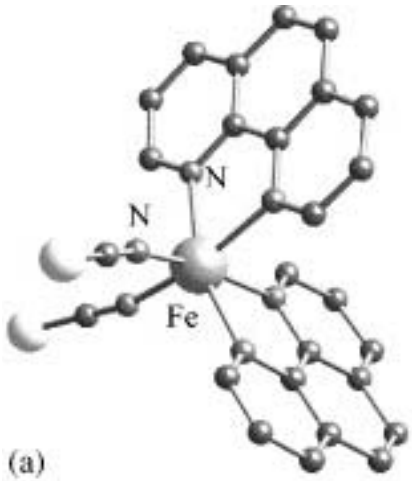


Strength of the ligand, type of complex , size of the metal and its oxidation state affects crystal field splitting

Temperature dependence of magnetic moments

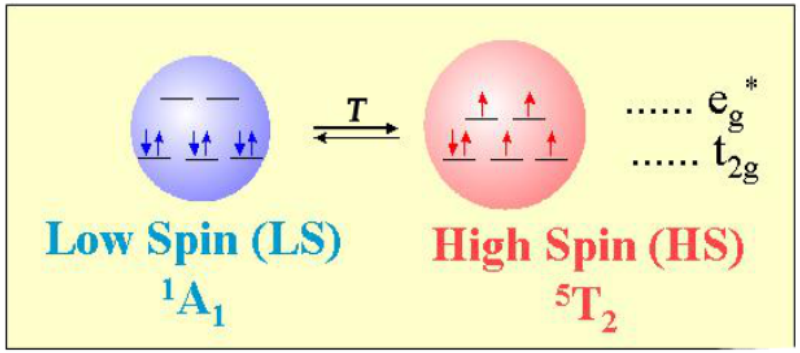


$Fe(Phen)_2(NCS)_2$



How crystal field theory explains temperature dependence of μ_{eff}

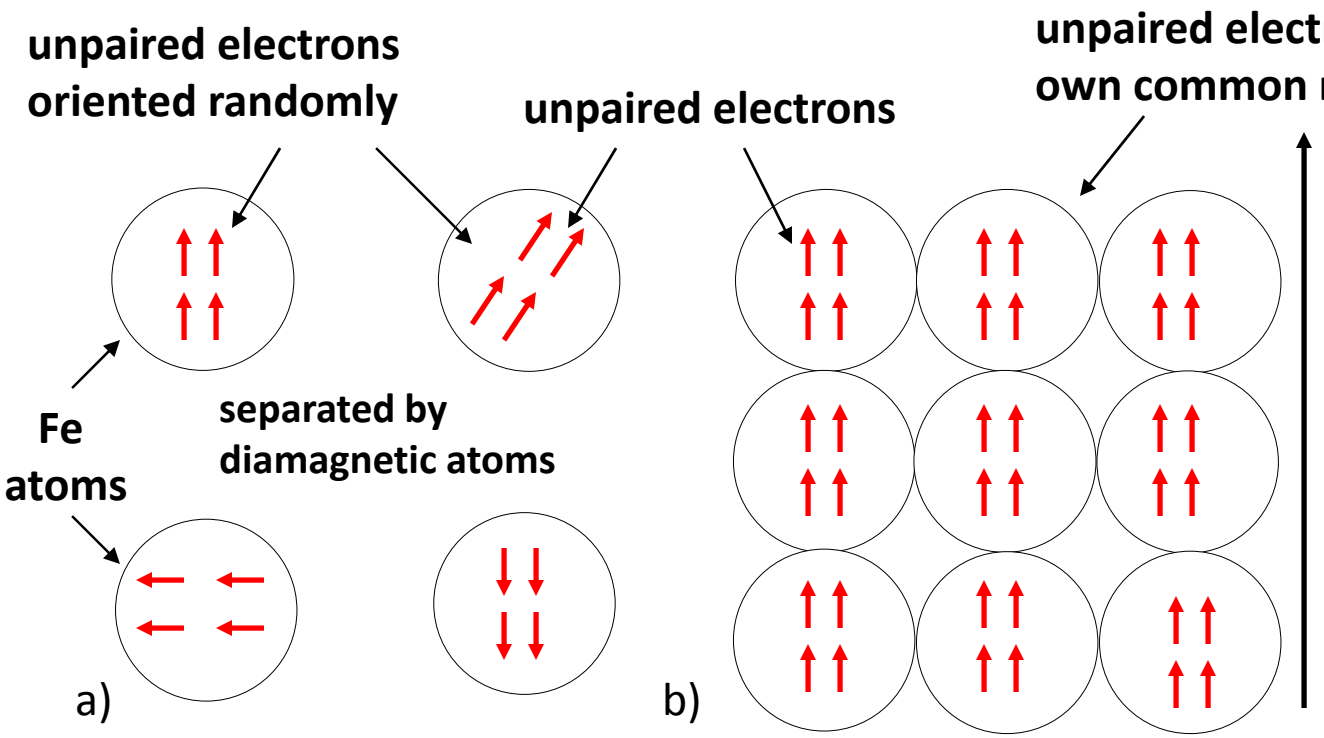
$Fe^{2+} = d^6$



Ferromagnetism:

In a normal paramagnetic material, the atoms containing the unpaired electrons are magnetically dilute, and so the unpaired electrons in one atom are not aligned with those in other atoms.

In ferromagnetic materials, such as metallic iron, or iron oxides [magnetite (Fe_3O_4)], where the paramagnetic iron atoms are very close together, they can create an internal magnetic field strong enough that all the centers remain aligned:

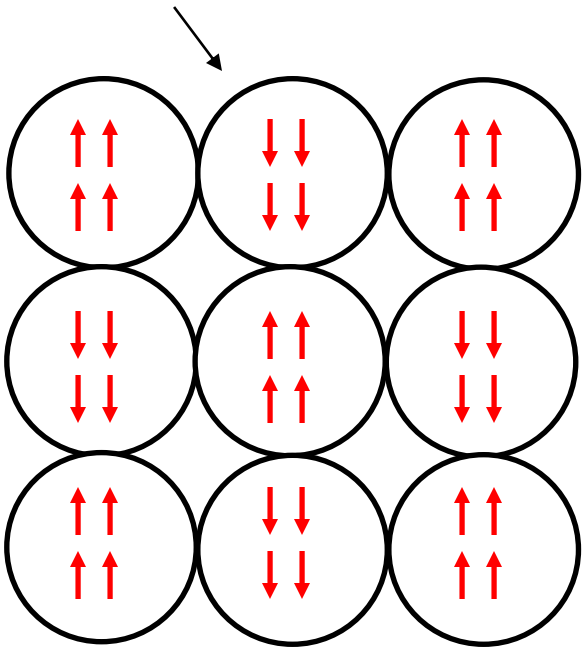


a) **paramagnetic,** magnetically dilute in e.g. $[Fe(H_2O)_6]Cl_2$.

b) **ferromagnetic**, as in metallic Fe or some Fe oxides.

Antiferromagnetism:

electron spins in opposite directions in alternate metal atoms

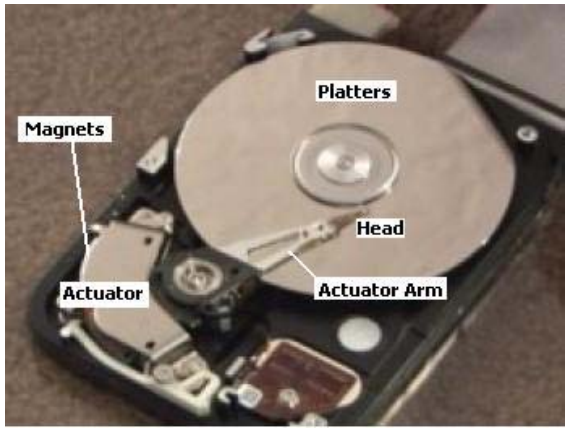


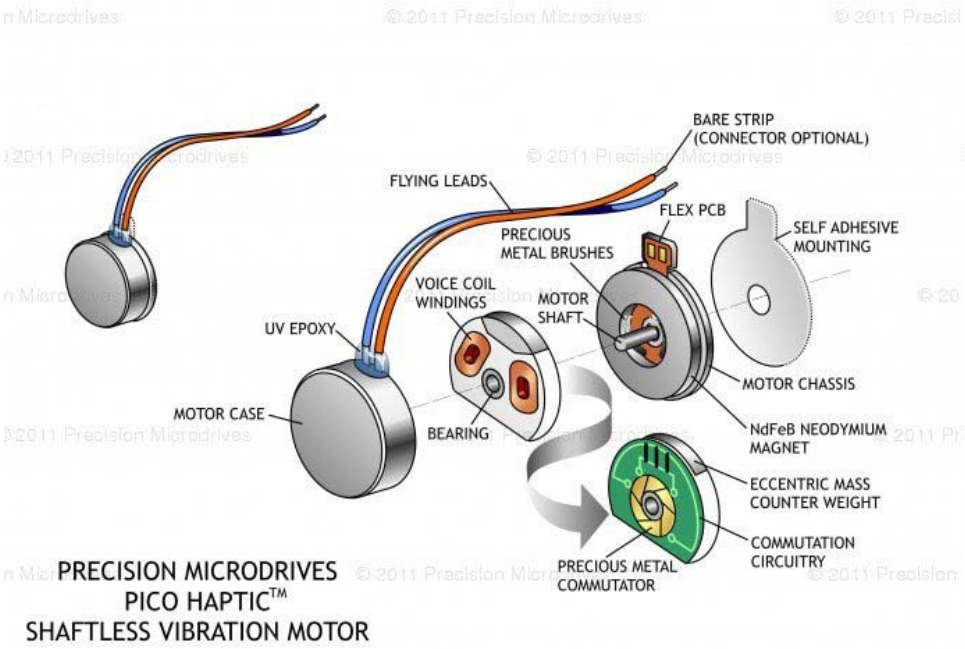
antiferromagnetism

Here the spins on the unpaired electrons become aligned in opposite directions so that the μ_{eff} approaches zero, in contrast to ferromagnetism, where μ_{eff} becomes very large. An example of anti-ferromagnetism is found in MnO and oxymyloglobin.

Observed in Oxyhemoglobin and oxymyoglobin

Which is the strongest ferromagnetic material in the world?





NEODYMIUM MAGNET

$Nd_2Fe_{14}B$

Problem solving !!

Cobalt form three octahedral complexes, $[MCl_6]^{4-}$, $[M(R_2NCS_2)_3]$ and $[M(CN)_5(H_2O)]^{3-}$. These have magnetic moments 0, 1.73 and 3.9 BM (not in the order). Assign the moments to the complexes and predict the relative strength of the dithiocarbamate ligand.

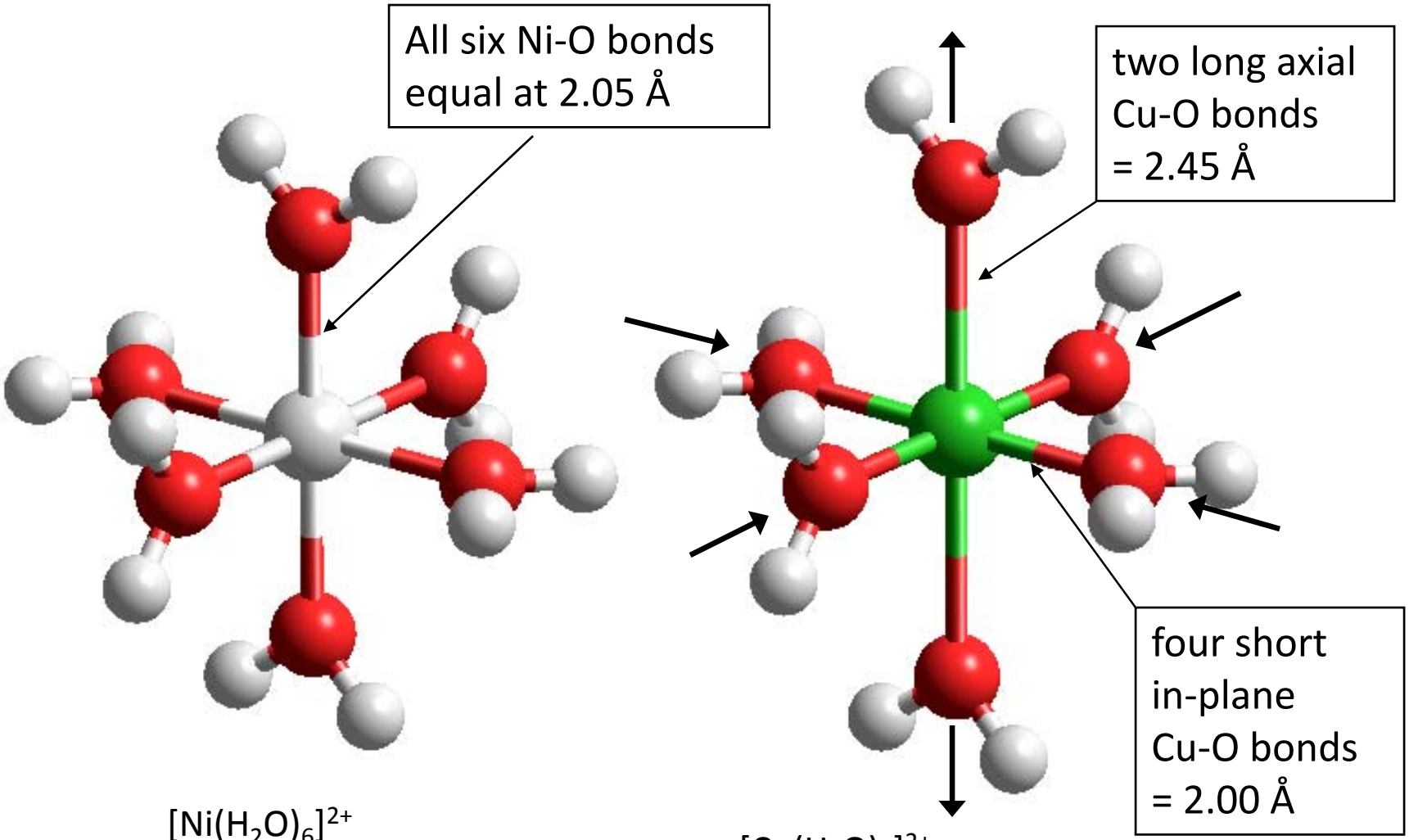
Solution

$[CoCl_6]^{4-}$ Co^{2+}, 3d ^{7} ,Cl- weak ligand, HS, 3 unpaired e's ~ 3.9 BM

$[Co(CN)_5(H_2O)]^{3-}$. Co^{2+}, CN- strong ligand, LS 1 unpaired e 1.73 BM

$[Co(R_2NCS_2)_3]$, Co^{3+}, 3d ^{6} , 0 BM means ligand is strong field and low spin complex

Distortion of some octahedral complexes

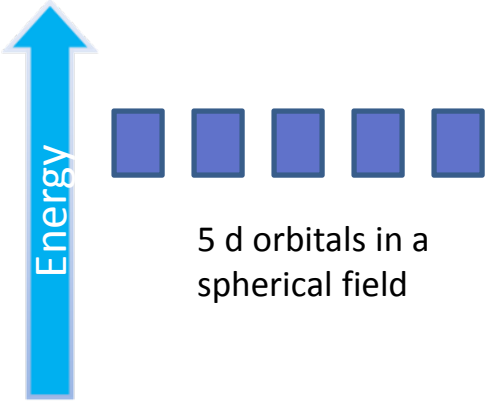


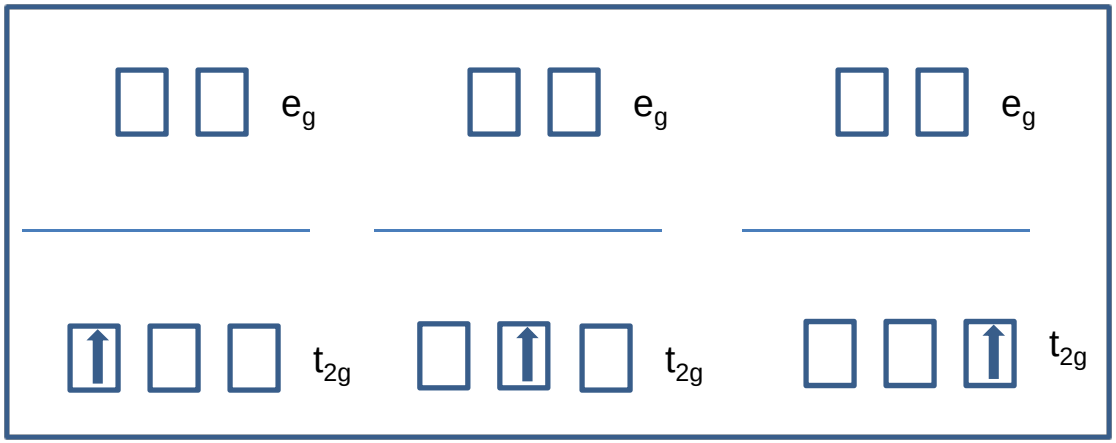
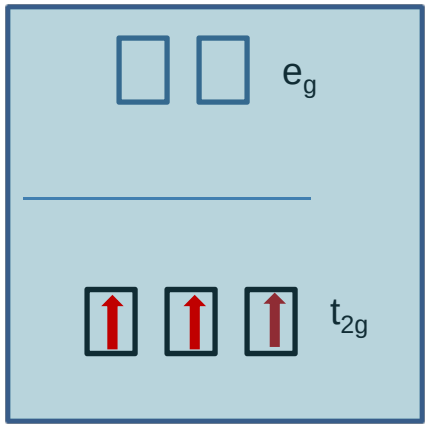
$[Ni(H_2O)_6]^{2+}$
no distortion

$[Cu(H_2O)_6]^{2+}$
distortion lengthens axial Cu-O's

Orbital Degeneracy versus Electronic Degeneracy

Orbital degeneracy: The 5 d orbitals are of the same energy in spherical field. This orbital degeneracy is lifted (reduced) in an octahedral field (to t_{2g} and e_g)



Electronic Degeneracy : Case of t_{2g}^1
Here there are three different ways by which the single electron can occupy the t_{2g} orbitals. There are three possible electronic configurations which are of the same energy. **Electronic degeneracy is present**

Electronic Degeneracy : Case of t_{2g}^3
Here there is only one way by which the three electron can occupy the t_{2g} orbitals. So there is **no electronic degeneracy**